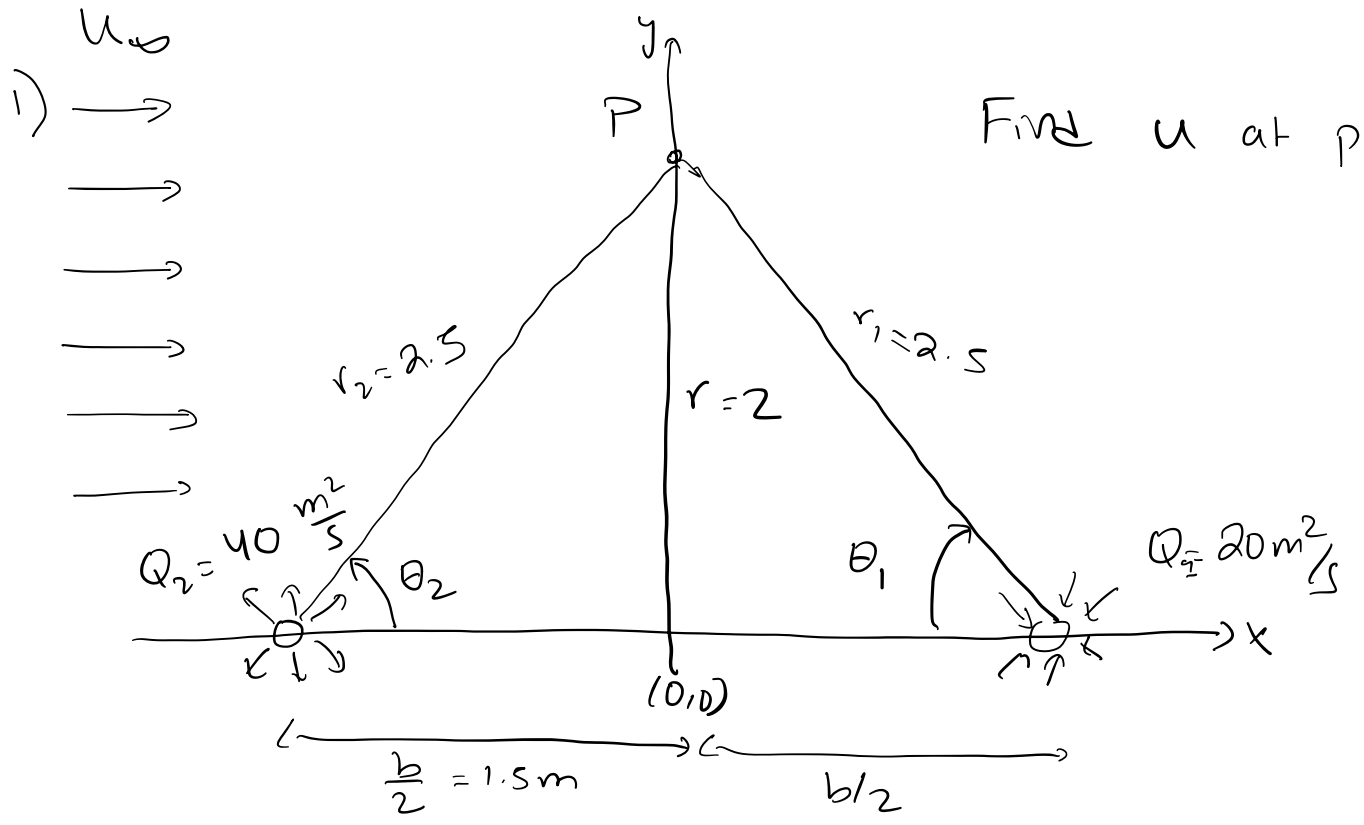


Find u at point 1

$$r^2 = r_1^2 + \left(\frac{b}{2}\right)^2 = 2m$$

Uniform flow: $\psi = U_\infty y$

Source flow: $\psi = \frac{\Lambda}{2\pi}\theta$

$$\psi = \psi_{source} + \psi_{sink} + \psi_{uniform}$$

$$\psi = \frac{\Lambda_2}{2\pi}\theta_2 - \frac{\Lambda_1}{2\pi}\theta_1 + U_\infty y$$

$$\theta_2 = \tan^{-1}\left(\frac{y}{x-(-1.5)}\right) = \tan^{-1}\left(\frac{y}{x+1.5}\right)$$

$$\theta_1 = \tan^{-1}\left(\frac{y}{x+1.5}\right)$$

$$\psi = \frac{\Lambda_2}{2\pi}\tan^{-1}\left(\frac{y}{x+1.5}\right) - \frac{\Lambda_1}{2\pi}\tan^{-1}\left(\frac{y}{x-1.5}\right) + U_\infty y$$

$$\Lambda_2 = 2\Lambda_2$$

$$\psi = \frac{\Lambda_1}{2\pi}\left[2\tan^{-1}\left(\frac{y}{x+1.5}\right) - \tan^{-1}\left(\frac{y}{x-1.5}\right) + U_\infty y\right]$$

$$u = \frac{\partial \psi}{\partial y} \qquad v = -\frac{\partial \psi}{\partial x}$$

$$v = \frac{\Lambda_1}{2\pi}\left[\frac{2y}{(x+1.5)^2+y^2} + \frac{y}{(x-1.5)^2+y^2}\right]$$

Plug in $y=2$, $x=0$ & $\Lambda_1 = 20$

$$\boxed{v = 1.019 \text{ m/s}}$$

2) Consider the non-lifting flow over a cylinder of a given radius, where $V_\infty = 20$ ft/s. If V_∞ is doubled, that is $V_\infty = 40$ ft/sec., does the shape of the streamlines change? Explain

Non-lifting cylinder = Uniform flow + doublet

$$\Psi = (V_\infty r \sin\theta)\left(1 - \frac{R^2}{r^2}\right)$$

$$V_r = \frac{1}{r}\frac{\partial \Psi}{\partial \theta} = \frac{1}{r}\left(V_\infty r \cos\theta\right)\left(1 - \frac{R^2}{r^2}\right)$$

$$\frac{V_r}{V_\infty} = \cos\theta\left(1 - \frac{R^2}{r^2}\right)$$

$$V_\theta = \frac{\partial \Psi}{\partial r} = V_\infty \sin\theta\left(r - \frac{R^2}{r}\right)$$

$$V_\theta = \frac{\partial \Psi}{\partial r} = V_\infty \sin\theta\left(1 + \frac{R^2}{r^2}\right)$$

$$\frac{V_\theta}{V_\infty} = \sin\theta\left(1 + \frac{R^2}{r^2}\right)$$

3) Consider a lifting flow over a cylinder of a given radius & w/ given circulation. If V_∞ is doubled and the circulation is kept the same, does the shape of the streamlines change

$$\Psi = \underbrace{(V_\infty r \sin\theta)\left(1 - \frac{R^2}{r^2}\right)}_{\text{Same as before}} + \underbrace{\frac{\Gamma}{2\pi} \ln\frac{r}{R}}_{\text{New term}}$$

We will only look at the new term

$$V_r = \frac{1}{r}\frac{\partial}{\partial\theta}\left(\frac{\Gamma}{2\pi}\ln\frac{r}{R}\right) = 0$$

$$V_\theta = -\frac{\partial}{\partial r}\left(\frac{\Gamma}{2\pi}\ln\frac{r}{R}\right) = -\frac{\partial}{\partial r}\left[\frac{\Gamma}{2\pi}(\ln r - \ln R)\right]$$

$$V_\theta = -\frac{\Gamma}{2\pi r}$$

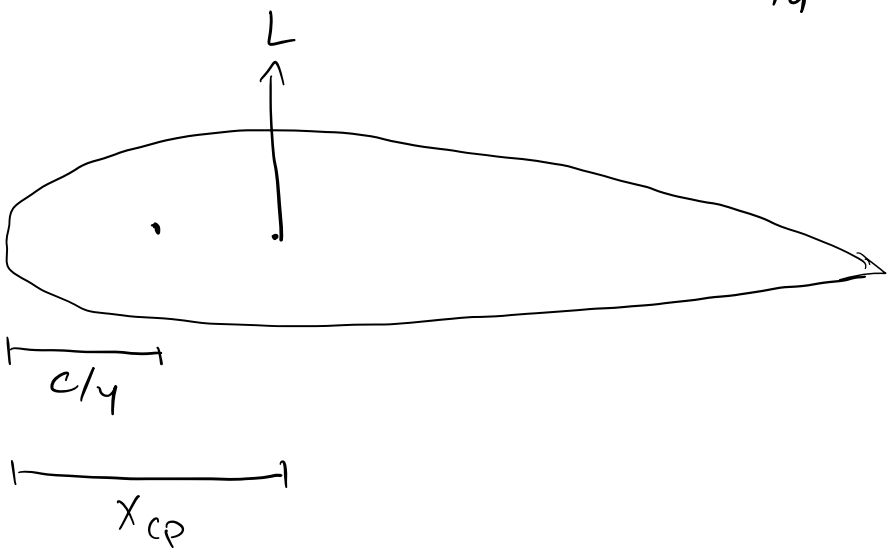
Bringing back old term & divide by V_∞

$$\frac{V_\theta}{V_\infty} = -\sin\theta\left(1 + \frac{R^2}{r^2}\right) - \frac{\Gamma}{2\pi r V_\infty}$$

Since V_∞ is on the right side if we change V_∞, the streamlines will change

4) In low speed, incompressible flow, an experiment showed

For $\alpha = 4^0 \rightarrow C_\ell = 0.85$ & $C_{M_{c/4}} = -0.09$. Find x_{cp}.



$$M'_{LE} = -\frac{c}{4} L' + M'_{c/4} = -x_{cp} L'$$

$$x_{cp} = \frac{c}{4} - \frac{M'_{c/4}}{L'}$$

$$\frac{x_{cp}}{c} = \frac{1}{4} - \frac{\left(M_{c/4} / q_\infty c^2\right)}{\left(L' / q_\infty c\right)}$$

$$= \frac{1}{4} - \frac{C_{m,c/4}}{C_\ell}$$

$$= \frac{1}{4} - \frac{(-0.09)}{0.85} = \boxed{0.356}$$

5) Consider a NACA 23012 airfoil. The mean camber line for this airfoil is given by

(1)
$$\frac{z}{c} = 2.6595\left[\left(\frac{x}{c}\right)^3 - 0.6075\left(\frac{x}{c}\right)^2 + 0.1147\left(\frac{x}{c}\right)\right]$$

for $0 \leq \frac{x}{c} \leq 0.2025$

(2)
$$\frac{z}{c} = 0.02208\left(1 - \frac{x}{c}\right) \quad \text{for} \quad 0.2025 \leq \frac{x}{c} \leq 1.0$$

Calculate (a) $\alpha_{L=0}$ b) C_l at $\alpha = 4^0$

$$\frac{dz}{d\left(\frac{x}{c}\right)}\left(\frac{z}{c}\right) = 2.6595\left[3\left(\frac{x}{c}\right)^2 - 2\cdot 0.6075\left(\frac{x}{c}\right) + 0.1147\right]$$

$$\frac{dz}{dx} = -0.02208$$

Transforming x to θ when $x = \frac{c}{2}(1-\cos\theta)$

(1)
$$\frac{dz}{dx} = 2.6595\left[3\left(\frac{\frac{c}{2}(1-\cos\theta)}{c}\right)^2 - 1.215\left(\frac{\frac{c}{2}(1-\cos\theta)}{c}\right) + 0.1147\right]$$

$$= 2.6595\left[\frac{3}{4}\left(1-2\cos\theta+\cos^2\theta\right)-0.6075\left(1-\cos\theta\right)+0.1147\right]$$

$$= 2.6595\left[\frac{3}{4}-\frac{3}{2}\cos\theta+\frac{3}{4}\cos^2\theta-0.6075+0.607\cos\theta+0.1147\right]$$

$$= 0.6840-0.2736\cos\theta+1.995\cos^2\theta$$

$$0.2025=\frac{c}{2}\left(1-\cos\theta\right)$$

$$\cos^{-1}\left(1-\frac{2(0.2025)}{c=2}\right)=0.9335 \text{ rads}$$

a) $$\alpha_{L=0}=-\frac{2}{\pi}\int_0^{\pi}\frac{dz}{dx}\left(\cos\theta-1\right)d\theta$$

$$\alpha_{L=0}=-\frac{1}{\pi}\int_0^{0.9335}\frac{dz}{dx}\left(\cos\theta-1\right)d\theta$$

$$=-\frac{1}{\pi}\int_0^{0.9335}\left(0.6840-2.3736\cos\theta+1.995\cos^2\theta\right]\left(\cos\theta-1\right)d\theta$$

$$-\frac{1}{\pi}\int_{0.9335}^{\pi}0.02208-0.02208\cos\theta\,d\theta$$

$$=-\frac{1}{\pi}\int_0^{0.9335}0.6840\cos\theta-2.3736\cos^2\theta+1.995\cos^3\theta$$

$$-\left[0.6840-2.3736\cos\theta+1.995\cos^2\theta\right]d\theta$$

$$-\frac{2}{\pi}\int_{0}^{0.9935} -0.6840 - 3.056\cos\theta - 4.3886\cos^2\theta + 1.995\cos^3\theta \, d\theta$$

From integral tables:

$$\int \cos\theta \, d\theta = \sin\theta \qquad \int \cos^2\theta \, d\theta = \frac{1}{2}\sin\theta\cos\theta + \frac{1}{2}\theta$$

$$\int \cos^3 d\theta = \frac{1}{3}\sin\theta\left(\cos^2\theta + 2\right)$$

$$\alpha_{2:0} = \frac{1}{\pi}\left[-0.6840 + 3.0576\sin\theta - 4.3686\left(\frac{1}{2}\sin\theta\cos\theta + \frac{1}{2}\theta\right) + 1.995\left[\frac{1}{3}\sin\theta\left(\cos^2\theta + 2\right)\right]\right]_{0}^{0.9335}$$

$$= -\frac{1}{\pi}\left[-0.6840 + 3.0576\sin\theta - 2.1843\sin\theta\cos\theta - 2.1843\theta + 0.665\sin\theta\left(\cos^2\theta + 2\right)\right]_{0}^{0.9335}$$

$$-\underbrace{\frac{1}{\pi}\left[0.02208\,\theta - 0.0228\sin\theta\right]_{0.9335}^{\pi}}$$

Second part

$$\alpha_{L=0} = -\frac{1}{\pi}\left(-0.0085 + 0.0065\right) = -0.0191 \text{ rad}$$

$$= \alpha_{L=0} = -1.09^{0}$$

b) $C_L = 2\pi(\alpha - \alpha_{L=0}) = 2\pi\left(0.06987 + 0.0191\right) =$

$$\boxed{0.559}$$

Example 4.8:

Consider an airfoil with $Re_c = 3.1 \times 10^6$. Estimate (a) the laminar boundary layer thickness at the trailing edge for a chord length of 1.5 m and (b) the net laminar skin-friction drag coefficient for the airfoil.

a) $\delta = \frac{5x}{\sqrt{Re_c}} = \frac{5(1.5)}{\sqrt{3.1 \times 10^6}} = 0.00426 \text{ m}$

b) $C_f = \frac{1.326}{\sqrt{Re_c}} = 7.54 \times 10^{-4} \cdot 2 = \boxed{0.0015}$

Example 4.9:

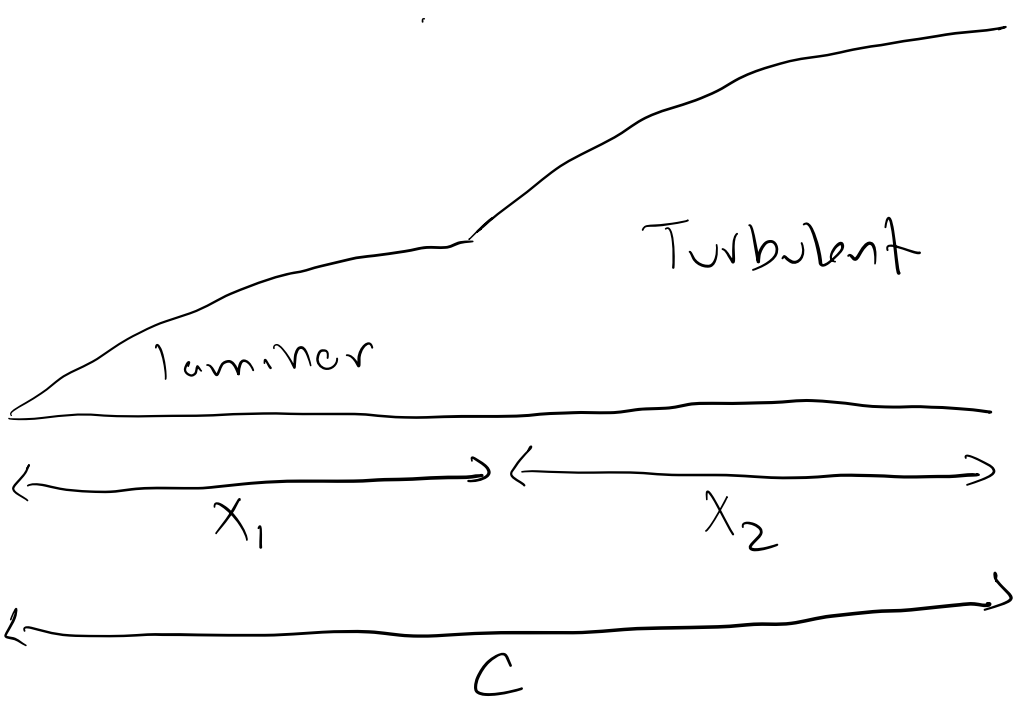
Consider an airfoil with $Re_c = 3.1 \times 10^6$. Estimate (a) the turbulent boundary layer thickness at the trailing edge for a chord length of 1.5 m and (b) the net turbulent skin-friction drag coefficient for the airfoil.

a) $\delta = \frac{0.37x}{Re_x^{1/5}} = \boxed{0.0279 \text{ m}}$

b) $C_F = \frac{0.074}{Re_c^{1/5}} = 0.00372 \cdot 2 = \boxed{0.00744}$

Example 4.10

Consider an airfoil with $Re_c = 3.1 \times 10^6$. Estimate the skin-friction drag but now assume the critical Reynolds number $Re_{cr} = 500$ K.



$$Re_{x_{cr}} = \frac{\rho_\infty V_\infty x_{cr}}{\mu_\infty} = 5 \times 10^5$$

$$x_{cr} = x_1$$

$$= \frac{\rho_\infty V_\infty x_1}{\mu_\infty} = 5 \times 10^5$$

$$Re_c = \frac{\rho_\infty V_\infty c}{\mu_\infty} = 3.1 \times 10^6$$

$$\frac{Re_{x_{cr}}}{Re_c} = \frac{5 \times 10^5}{3.1 \times 10^6} = 0.1613 \simeq \frac{x_1}{c}$$

turb. = turbulent

lam. = laminar

$(D_{f.c})_{turbulent}$

- $D_{f.c}$: length region being considered
- turbulent: State of boundary layer

$$(D_{f.c})_{turb} = q_\infty S (C_{f.c})_{turb.}$$

$$S = \text{Area} \simeq b \times c$$

(b: span)

Assume span = 1

Turbulent drag on whole chord is:

$$(D_{f.c})_{turb.} = q_\infty (1) c (C_{f.c})_{turbulent}$$

Turbulent drag on just region 1 is:

$$(D_{f,c})_{turb} = q_\infty X_1 (C_{f,2})_{turb}$$

Turbulent drag on just region 2 is:

$$(D_{f,2})_{turb.} = (D_{f,c})_{turb} - (D_{f,1})_{turb}$$

$$(D_{f,2})_{turb} = q_\infty C (C_{f,c})_{turb} - q_\infty X_1 (C_{f,2})_{turb}$$

Laminar drag on region 1 is $(D_{f,2})_{laminar}$

$$(D_{F,1})_{laminar} = q_\infty X_1 (C_{f,1})_{laminar}$$

$$D_f = (D_{f,1})_{laminar} + (D_{f,2})_{turbulent}$$

$$D_f = q_\infty X_1 (C_{f,1})_{laminr} + q_\infty C (C_{f,c})_{turb} - q_\infty X_1 (C_{f,2})_{turb}$$

$$C_f = \frac{D_f}{q_\infty S} = \frac{D_f}{q_\infty (1) C}$$

$$C_f = \frac{X_1}{C} (C_{f,1})_{lam.} + (C_{f,2})_{turb.} - \frac{X_1}{C} (C_{f,2})_{turb.}$$

$$C_f = 0.1613 (C_{f,1})_{lam} + (C_{f,c})_{turb} - 0.1613 (C_{f,2})_{turb}$$

$$Re_{x_1} = \frac{x_1}{c} Re_c = 0.1613\,(3.1\times10^6) = 5\times10^5$$

$$(C_{f,c})_{lam.} = \frac{1.328}{\sqrt{Re_{x_1}}} = \frac{1.328}{\sqrt{5\times10^5}} = 0.00188$$

$(C_{f,turb}) \longrightarrow$ From previous example

$= 0.00372$

$$(C_{f,1})_{turb} = \frac{0.074}{Re_x^{1/5}} = \frac{0.074}{(5\times10^5)^{1/5}} = 0.00536$$

$$C_F = 0.1613\,(0.00188) + 0.00372 - 0.1613\,(0.00536)$$

$$= 0.003158$$

Net $C_F = 2(0.003158) = \boxed{0.0063}$